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Detection of Cyber Harassers on Social Media: Progress Report One

This idea is born out of a pressing need to curb a growing problem. Even the best of things come with their pitfalls. Growing technology and connectivity have led to an increased number of child predators and cyber stalkers online. My solution aims to efficiently nip this problem in the bud using advanced Machine Learning techniques such as Support Vector Machines (SVM), Decision Trees, K-Nearest Neighbours and Random Forests.

The use cases for this problem are virtually unlimited. Users on almost all social media websites and discussion forums can fall prey to such predators. This tool will be prototyped on a particular platform and then generalized to suit the requirements of all such platforms, thereby leaving a huge population much safer than they were before. Any website can use this tool to flag suspicious profiles and become a much better space for all its users.

Social media has opened doors for people who misuse it to undermine, harass, humiliate. Threaten and bully others. There is a lack of resources to detect and hinder its occurrence. On average, about 16% students have admitted that they have been bullied at some point in their lives, as per surveys conducted in the United States. The problem is hence indeed grave and needs to be addressed.

The dataset I have used for this project is the one provided at Kaggle, which was part of a contest held 8 years ago. https://www.kaggle.com/c/detecting-insults-in-social-commentary. There is a separate training data set with roughly 4000 tuples and a testing data set with about 2500 tuples.

Each entry contains a date, a comment and a label, where ‘1’ indicates that it is an insult and ‘0’ indicates that it is neutral.

	A	B	C
1	Insult	Date	Comment
2	1	20120618	"You fuck your dad."
3	0	20120528	"i really don't understand your point.\xa0 It seems that you are mixing apples and oranges."
4	0		"A\\xc2\\xa0majority of Canadians can and has been wrong before now and will be again.\\n\\nUnless you're supportive of the idea that nothing is full proof or perfec
5	0		"listen if you dont wanna get married to a man or a women DONT DO IT. what would it bother you if gay people got married stay in your lane do you let them do them.
6	0	20120619	"C\xe1c b\u1ea1n xu\u1ed1ng \u0111\u01b0\u1eddng bi\u1ec3u t\xecnh 2011 c\xf3 \xf4n ho\xe0 kh\xf4ng ? \nC\xe1c ng\u01b0 d\xe2n ng\u1ed3i cu\xed \u0111\u
7	0	20120620	"@SDL OK, but I would hope they'd sign him to a one-year contract to start with. Give him the chance to be reliable and productive, but give themselves the out if all hi
8	0	20120503	"Yeah and where are you now?"
9	1		"shut the fuck up. you and the rest of your faggot friends should be burned at the stake"
10	1	20120502	"Either you are fake or extremely stupid...maybe both..."
11	1	20120620	"That you are an idiot who understands neither taxation nor women's health."
12	0	20120620	"@jdstorm dont wish him injury but it happened on its OWN and i DOUBT he's injured, he looked embarrassed to me"
13	0	20120530	"Be careful,Jimbo.OG has a fork with your name on it."
14	0		"@tonnyb Or they just don't pay attention "
15	0		"Hmmm. Perhaps some who are too pig-faced to get laid and therefore have zero chance of getting pregnant from such activity hold something against women who ca
16	0		"@barrettmarson Huh? Her income was $21,912. Her spending was $5,470. My math puts that at 24.9%."
17	1	20120611	"FOR SOME REASON U SOUND RETARDED. LOL. DAMN. WHERE U BEEN NEGRO"
18	1	20120320	"You with the 'racist' screen name\n\nYou are a PieceOfShit....."
19	0		"Oh! & cheating as well...1966 cup was a cheat fest, & you still dare to post that comment..shame on you lol"
20	1	20120320	"your such a dickhead..."
21	1	20120320	"Your a retard go post your head up your #%&*"
22	0	20120612	"@EephusBlue\xa0Makes you want to say "Mike MacWHOgal?""
23	0	20120611	"http://www.youtube.com/watch?v=tLYLLPHKRU4"
24	0	20120502	"And you know they've burned holes in all the carpeting."

Fig. 1. A snippet of the training data set

The testing data set contains an additional column called ‘Usage’, however it is not relevant to our project at the moment.

	A	B	C	D	E
1	Insult	Date	Comment	Usage	
2	0		"THE DRUDGE REPORT\\n\\n\\n\\nYou won't see this story on foxfag forum because they suck blacks and gay 0bam	PublicTest	
3	0	20120618	"@ian21\xa0"Roger Clemens is the fucking man, and never did any fucking steroids because he is fucking awesome.	PublicTest	
4	1	20120618	"Agree with Alan you are an extremest idiot. You are not American, we native americans are, you are an alien"	PublicTest	
5	0		"Really?\\n\\nI see Marc Lamont Hill on various programs quite often, as is Al Sharpton, and Michael Moore even g	PrivateTest	
6	0	20120620	"Really suck isn't the word, when many of our nuclear power plants start melting down, it will literally be hell on ear	PrivateTest	
7	0		"...after a couple guys like him rape him first...."	PrivateTest	
8	0	20120529	"Matthews mentored Perry at USC. Bodes well..."	PrivateTest	
9	0	20120619	"How about we accept facts and patiently wait for you to back up your claim."	PrivateTest	
10	1	20120609	"To engage in an intelligent debate with you is like debating to a retarded person. It's useless. It looks like you're be	PublicTest	
11	0	20120612	"Same reason no one cared Obama did cocaine, which is a felony as well."	PublicTest	
12	0	20120610	"Well, Kris, you might just be surprised. Can you say Reagan Democrats? ROMNEY has a grasp of reality and the ECO	PrivateTest	
13	0	20120610	"In reality GWB put more people on food stamps because of his 100% failed economic policies!"	PrivateTest	
14	0		"because the zionists will blow us up if we don't at least try!"	PrivateTest	
15	0	20120528	"Thank you. Unfortunately on this kindle I'm limited on how far back I can go."	PrivateTest	
16	0	20120610	"CNN supports HARCORE MASS MURDERING ISLAMIST TERRORISTS!!!!\n\n\n\nDon\u2019t ever forget this guys!	PublicTest	
17	0	20120529	"What is your problem AW"	PrivateTest	
18	0	20120529	"Name a corporation that has told the truth."	PrivateTest	
19	0	20120528	"You know, I don't think I ever actually put my gloves in the glove box..."	PrivateTest	
20	0	20120529	"then they shouldnt get mad when they read it."	PrivateTest	
21	0	20120612	"Richards passes it to Carter. \xa0Wow. \xa0Surprising"	PrivateTest	
22	0		"I\\'m a second generation German working on my Master\\'s \\xc2\\xa0so I doubt you would be accurate in saying	PublicTest	
23	0		"@Fenestrate Even tough Mitt doesn't drink. I thought about Mitt Sat nite when I tried get into Slippery Noodle bar	PrivateTest	
24	0	20120528	"Then you don't understand basketball."	PrivateTest	
25	1	20120320	"GO SUCK_YOU DADDYS_COCK_OBAMA LOVING LIBERAL_FAGGOT"	PrivateTest	
26	0	20120619	"yeah!!!! fuck undying!"	PrivateTest	
27	0		"Circumstances under which abortion is legal in Israel:\\nThe termination committee approves abortions, u	PrivateTest	

Fig. 2. A snippet of the testing data set

```
In [6]:  #View dataset schema
         df.head()
```

Out[6]:

	Insult	Date	Comment
0	1	20120618192155Z	"You fuck your dad."
1	0	20120528192215Z	"i really don't understand your point.\xa0 It ...
2	0	NaN	"A\\xc2\\xa0majority of Canadians can and has ...
3	0	NaN	"listen if you dont wanna get married to a man...
4	0	20120619094753Z	"C\xe1c b\u1ea1n xu\u1ed1ng \u0111\u01b0\u1edd...

```
In [7]:  #Remove irrelevant features
         df = df.drop('Date',axis=1)
```

```
In [56]:  #Check for null values
          import seaborn as sb
          import matplotlib
          sb.heatmap(df.isnull())
```

```
Out[56]: <matplotlib.axes._subplots.AxesSubplot at 0x2339723cdc8>
```

Fig. 3. I remove irrelevant attributes from the training data set and check for duplicates, the same is also done for the testing data set.

Now, since we have just one feature that is to be used to predict the label, I start to build a vocabulary, using somewhat a bag of words approach. It is important to note here that this data set is off a website that has casual chats and hence, there are typos galore, The data is pretty dirty and preprocessing and cleaning takes up a major chunk of the effort.

I saw a large number of words with front slashes and backslashes like \xe2, which are probably escape sequences and have a special meaning, so to remove them, as they are garbage to us, I create a regular expression of the form:

slash_all = r'\s(?:[\w_]*[/\\](?:[\w_]*[/\\])*[\w_]*)'*

Which is then followed by:

comment = re.sub(slash_all,' ',comment)

I also remove all special characters except the apostrophe (‘), trim extra whitespace and change every character to lower case. I then tokenise and remove stop words. After this, stemming is done as the last stage. Collectively, these steps work to reduce the size of the vocabulary and prevent it from blowing up.

```
comments_test = []
for i in range(df2.shape[0]):
    comment = df2.iloc[i, 1]
    #Regular expression for words with slashes
    slash_all = r'\s*(?:[\w_]*[/\\](?:[\w_]*[/\\])*[\w_]*)'
    #Remove words with slashes
    comment = re.sub(slash_all,' ',comment)
    #Remove special characters except '
    comment = re.sub('[^\'A-Za-z]',' ',comment)
    #Remove extra whitespace
    comment = comment.strip()
    #Change to lower case
    comment = comment.lower()
    #Tokenise
    tokenised_comment = wt(comment)
    #Remove stop words
    comments_processed = []
    for word in tokenised_comment:
        if word not in set(stopwords.words('english')):
            comments_processed.append(stemmer.stem(word))
    comment_text = " ".join(comments_processed)
    comments_test.append(comment_text)
```

Fig. 4. The word manipulation code

Then, the feature matrix is created, and the training and testing data sets are transformed.

```
# Create feature matrix
from sklearn.feature_extraction.text import CountVectorizer
matrix = CountVectorizer()
```

```
X_train = matrix.fit_transform(comments).toarray()
y_train = df.iloc[:, 0]
X_test = matrix.transform(comments_test).toarray()
y_test = df2.iloc[:, 0]
```

Fig. 5. The set-up

Once the data is ready to go, I run it through three classifiers – Naïve Bayes, Decision Trees and Suppurt Vector Classifiers. The results of all three, in order, are shown below.

```
print(accuracy)
```

```
0.6422364941443143
```

```
print(cm)
```

```
[[1296  658]
 [ 289  404]]
```

```
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.82      0.66      0.73      1954
           1       0.38      0.58      0.46       693

    accuracy                           0.64      2647
   macro avg       0.60      0.62      0.60      2647
weighted avg       0.70      0.64      0.66      2647
```

Fig. 6. The accuracy, confusion matrix and classification report for the Naïve Bayes classifier

```
print(accuracy)
```

```
0.7842840952021156
```

```
print(cm)
```

```
[[1877   77]
 [ 494  199]]
```

```
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.79      0.96      0.87      1954
           1       0.72      0.29      0.41       693

    accuracy                           0.78      2647
   macro avg       0.76      0.62      0.64      2647
weighted avg       0.77      0.78      0.75      2647
```

Fig. 7. The accuracy, confusion matrix and classification report for the Decision Tree classifier

```
print(accuracy)
```

```
0.8111069134869664
```

```
print(cm)
```

```
[[1896   58]
 [ 442  251]]
```

```
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.81      0.97      0.88      1954
           1       0.81      0.36      0.50       693

    accuracy                           0.81      2647
   macro avg       0.81      0.67      0.69      2647
weighted avg       0.81      0.81      0.78      2647
```

Fig. 8. The accuracy, confusion matrix and classification report for the Support Vector classifier

Clearly, SVC gives the best results and an accuracy of 81% is pretty good. It is followed by Decision Tree and then by Naïve Bayes, which does not give a good result. This research can also be expanded to other datasets available on the internet, such as BayzickBullyingData, TextMiningAndCyberCrimeData, Formspring Data. I am also keen to use the random forest classifier on this as well as clustering. The Wikipedia list of bad words could also be added to the vocabulary to enhance the classification and we could also include TF-IDF to get a better result.

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